

Assignment No. 2

Rajdhani

DATE / /

Name: Thorat Arpita Ramchandra

Roll no: 114

Class: MCA-I

Subject: Data communication & network

Q1 Explain OSI Model in detail

- - OSI stands for open system Interconnection.
- It is a 7-layer architecture with each layer having specific functionality to perform
- All these 7 layers work collaboratively to transmit the data from one person to another across the globe.
- OSI model was developed by ISO - 'International organization for standardization', in year 1984.
- OSI model divides the whole task into 7 smaller and manageable tasks. Each layer is assigned a particular task.
- Each layer is self-contained, so that task assigned to each layer can be performed independently.

Layer - 7	Application	} Upper Layer or Software Layer
Layer - 6	Presentation	
Layer - 5	Session	
Layer - 4	Transport	} Heart of OSI
Layer - 3	Network	
Layer - 2	Data Link	} Lower Layer or Hardware Layer
Layer - 1	Physical	

OSI Model

1 Physical Layer:

- The lowest layer of the OSI reference model is the physical layer.
- It is responsible for the actual physical connection between the device
- The physical layer contains information in the form of bits
- Physical layer responsible for transmitting individual bits from one node to the next.
- When receiving data, this layer will get the signal received & convert into 0s & 1s and send them to the Data Link layer, which will put the frame back together
- Common physical layer devices are Hub, Repeater, Modem and cables.

2 Data Link layer:

- The data link layer is responsible for the node-to-node delivery of the message.
- The main function of this layer is to make sure data transfer is error-free from one node to another, over the physical layer.
- When a packet arrives in network, it is the responsibility of the DPL to transmit it to the Host using its MAC address.
- Packet in the Data link layer is referred to as frame
- Switches & Bridges are common Data Link layer devices
- The packet received from the network layer

is further divided into frames depending on the frame size of the NIC (network interface card)

- DLL encapsulates sender & receiver's MAC address in the header.

3 Network Layer:

- The Network Layer works for the transmission of data from one host to the other located in different networks
- It also takes care of packet routing i.e. selection of the shortest path to transmit the packet, from the number of routes available.
- The sender & receiver's IP address are placed in the header by the network layer.
- Segment in the Network layer is referred to as packet
- Network layer is implemented by networking devices such as routers & switches.

4 Transport Layer:

- The Transport Layer provides services to the application layer & takes services from the network layer
- It is responsible for the end-to-end delivery of the complete message.
- The transport layer also provides the acknowledgement of the successful data transmission and re-transmits the data if an error is found.
- Protocols used in Transport layer are TCP, UDP, NetBIOS
- It also adds Source & Destination port number in its header & forwards the segmented data to the network layers

- Generally, this destination port number is configured, either by default or manually.

5 Session Layer:

- The session layer is layer 5 layer from the bottom in the OSI model.
- The job of the session layer is to control & maintain connections between systems to share data.
- It establishes, maintains & ends sessions across all channels.
- In case of a network error, it checks the authenticity & provides recovery options for active sessions.
- It manages sessions & synchronizes data flow.

6 Presentation Layer:

- This layer is also known as translation layer, as this layer serves as a data translator for the network.
- The data which this layer receives from the application layer is extracted & manipulated here as per the required format to transmit over the network.
- The main responsibility of this layer is to provide or define the data format & encryption.
- The presentation layer is also called as syntax layer since it is responsible for maintaining the proper syntax of the data which it either receives or transmits to other layers.

7 Application Layer:

- This layer is analogous to the transport layer of the OSI model
- It is responsible for end-to-end communication & error-free delivery of data
- It shields the upper layer applications from the complexities of data.

2 Compare between OSI and TCP model

OSI Model	TCP/IP model
1 7 layers: Physical, Data link, Network, Transport, session, Presentation, Application	5 layers: Physical, Data link, Network, Transport, Application.
2 Separate session layer manages connections / Synchronization	No separate session layer, handled in Application layer
3 Presentation layer handles data formatting / encryption	Data formatting / encryption done within Application layer.
4 Theoretical model developed by ISO for standardization	Practical model developed by DoD for Internet use
5 Each layer strictly independent with clear interfaces	Layers more integrated less strict boundaries

6	No specific protocols defined, just functions	Define specific protocols: TCP, UDP, IP, HTTP etc
7	Used as reference for teaching / network design	Actual protocol stack used by Internet globally
8	Developed before widespread Internet use	Developed specifically for ARPANET / Internet Implementation

3 Explain in HTTP, FTP, SMTP, DNS, Telnet protocols of application layer.

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1 HTTP:

- HTTP stands for Hyper Text Transfer Protocol
- It is the foundation of the world wide web (www)
- HTTP works on the client server model
- This protocol is used for transmitting hypermedia documents like HTML
- This protocol was designed particularly for the communications betⁿ the web browser & web server, but this protocol can also be used for several other purposes
- HTTP is a stateless protocol (network protocol in which a client sends request to server & server responses back as per the given state), which means the server is not responsible for maintaining the previous client req^{ts}
- HTTP uses port number 80

2 FTP

- FTP Stands for file transfer protocol.
- This protocol helps to transfer different files from one device to another
- FTP Promotes sharing of files via remote computer devices with reliable, efficient data transfer
- FTP uses port number 20 for data access & port number 21 for data control.

3 SMTP

- Used for sending emails across networks
- works with mail transfer Agents (MTA) for message forwarding
- Port : 25 (SMTP), also 587 for secure transmission

4 DNS:

- DNS Stands for Domain name system
- The DNS service translates the domain name (selected by user) into the corresponding IP address
- For example, If you choose the domain name as www.abcd.com, then DNS must translate it as 192.36.20.8 (random IP address written just for understanding purpose)
- DNS protocol uses the port number 53

5 TELNET:

- Telenet Stands for telecommunications network
- This protocol is used for managing files over the Internet

- It allow the Telnet clients to access the resources of Telnet server
- Telnet Uses the port number 23

4 Explain services of Presentation layer.

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1 Translation:

- Data is sent from sender to receiver, but what if the sender device & receiver device understand different formats of code
- For ex, Suppose one device understands ASCII code & another device understands EBCDIC code
- In that, case the data must be translated into code that the recipient understands to determine what data has been sent
- It presentation layer is responsible for translating ASCII codes to EBCDIC or vice versa.
- With the help of the presentation layer the receiver understands the data effectively & uses it efficiently.

2 Encryption and Decryption:

- Whatever data is being transmitted betⁿ the sender & the receiver, that data must be secure
- Hackers can modify the data & send the modified data to the receiver to create false communication.

- The presentation layer is responsible for encrypting & decrypting data to avoid data leakage & data modification
- The plaintext data at the source is encrypted into ciphertext (unreadable format), then it is sent to the receiver, where the ciphertext is decrypted into plaintext
- Now if the hacker tries to hack the data, the hacker receives an encrypted, unreadable form & if the hacker tries to send modified data the receiver can detect the modification during decryption; thereby the data remains safe.

③ Compression & Decompression:

- If the file size is large, it becomes difficult to transmit the large file over the network.
- File size can be decreased by compressing the file for easy transmission of data
- Compression is the method of diminishing the size of a file to transmit data easily in less time
- When the compressed data reaches the receiver, the data is reconstructed back to the original size & this process is called decompression.

5 Explain Services of Session layer.
→ The session layer (layer 5) establishes, manages, maintains & terminates connections between applications on different network devices.

1 Session Establishment & Management:

- The layer initiates, maintains & terminates connections betⁿ applications, managing login session setup and ensuring proper disconnection

2 Synchronization & checkpointing:

- Inserts checkpoints into large data streams, allowing sessions to resume for the last checkpoint rather than restarting from the beginning if a failure occurs.

3 Activity Management:

- Manages the multiple, interruptible dialogues (activities) within a single session, allowing them to be interrupted & resumed explicitly

4 Dialogue control:

- Manages the "conversation" betⁿ computers by deciding who sends data & when, operating in either half-duplex (one at a time) or full duplex (simultaneous) modes

5 Session Termination:

- Gracefully terminates the session after data exchange is complete

6 Remote Procedure call (RPC):

- It supports RPCs, allowing programs to execute procedures on remote systems.

7 Authentication & Authorization:

- The session layer verifies user credentials (authentication) & verifies permissions (authorization) before allowing a session to connect to a service

6 Explain protocol & standards in detail

→ Protocol:

- Protocols define the format and order of messages exchanged betⁿ devices
- They also specify how devices should respond to various conditions.

• Common protocols include:

1 TCP/IP:

- This is the foundational protocol suite for the internet which governs how data is sent and received
- TCP ensures reliable transmission, while IP handles addressing & routing

2 HTTP:

- Used for transferring web pages on the internet
- It defines how messages are formatted & transmitted.

3 FTP:

- Used for transferring files between computers on a network
- It specifies how files are uploaded & downloaded.

4 SMTP:

- Used for sending emails
- It defines how emails messages are sent from one server to another

• Key components of protocol:

Protocols generally consist of three main elements:

- 1 Syntax: Defines the data structure, format, & signal levels (e.g., how the first 8 bits indicate the sender)
- 2 Semantics: Defines the meaning of each section of bits & how to interpret them (e.g., error handling or routing information)
- 3 Timing: Governs when data should be sent and at what speed to prevent congestion

• Standards:

- Standards are the set of rules for data communication that are needed for the exchange of information among devices.
- It is important to follow standards which are created by various standard organizations like IEEE, ISO, ANSI etc.

• Types of standards:

1 De Facto Standard:

- The meaning of the word "De Facto" is "By Fact" or "By convention".
- These are the standards that have not been approved by any organization but have been adopted as standard because of their widespread use.

• For example,

Apple & Google are two companies that established their own rules for their products which are different, Also they use some same standard rules for manufacturing their products.

2 De Jure Standard:

- The meaning of the word "De Jure" is "By law" or "By Regulations".
- Thus, these are the standards that have been approved by officially recognized bodies like ANSI, ISO, IEEE etc.
- These are the standards that are important to follow if it is required or needed.

For example,

All the data communication standard protocols like SMTP, TCP, IP, UDP etc. are important to follow the same when we need them.

• Key standards organizations include:

1. IEEE (Institute of Electrical & Electronics Engineers)
 - Develops standard for networking technologies, such as Ethernet (IEEE 802.3) & Wi-Fi (IEEE 802.11)
2. IETF (Internet Engineering Task Force):
 - Responsible for developing & promoting voluntary internet standards, particularly those related to TCP/IP.
3. W3C (World wide web consortium):
 - Develops standards for web technologies, including HTML & CSS.